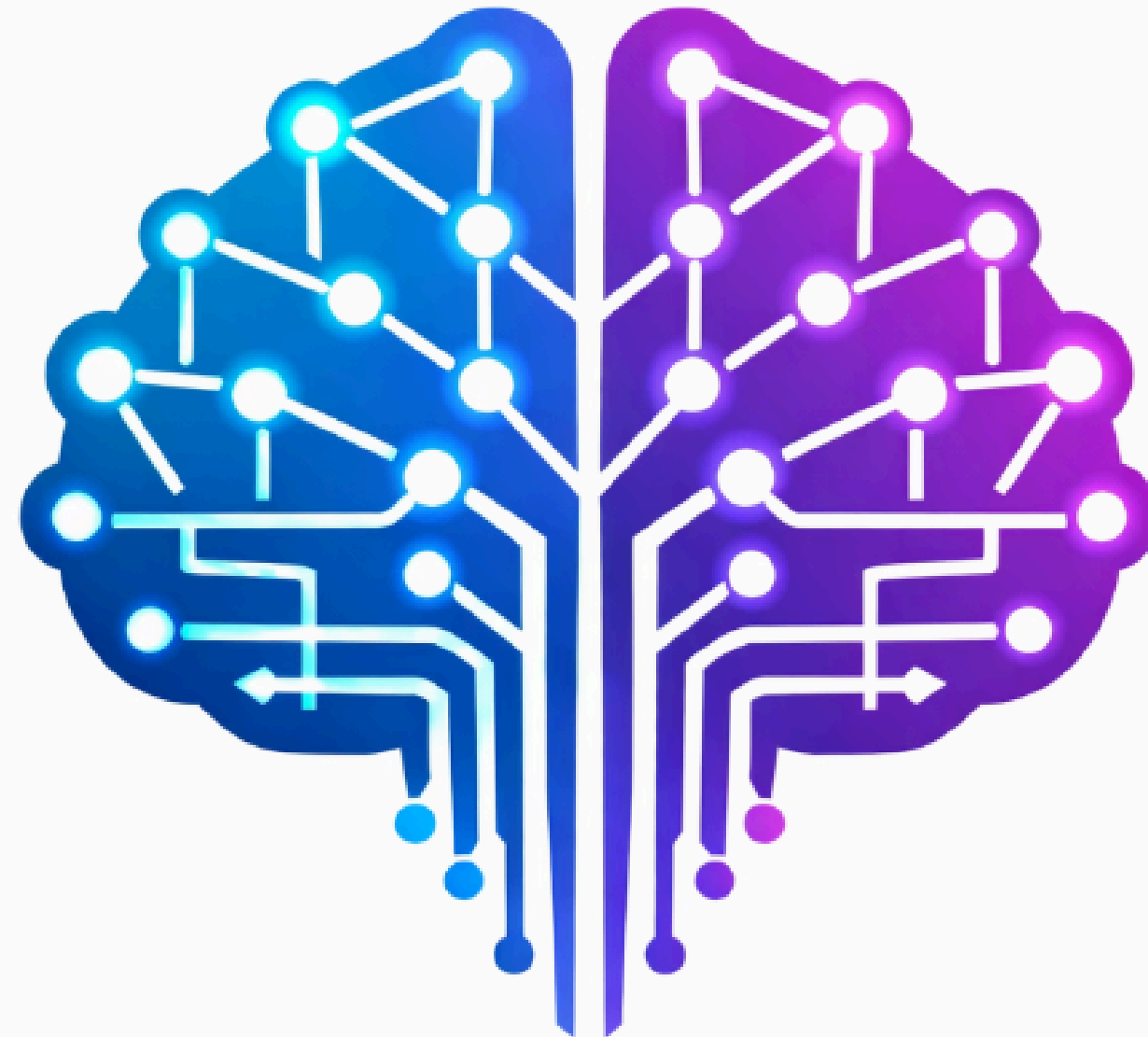


GENAI FUNDAMENTALS



LECTURE:
FINE-TUNING AT A HIGH LEVEL
(LORA & QLORA INTUITION)

Why Fine-Tuning Was Traditionally Difficult

Fine-tuning large language models used to be extremely expensive.

Challenges included:

- billions of model parameters
- large GPU requirements
- high training cost
- long training time

Updating the entire model was often impractical.

The Key Idea Behind Efficient Fine-Tuning

Instead of updating the entire model, modern methods **update small additional components**. The **main model stays frozen**.

Only **small layers** are trained to adapt model behavior.
This approach dramatically **reduces training cost**.

What Is LoRA?

LoRA stands for **Low-Rank Adaptation**.

Instead of modifying large weight matrices, LoRA adds **small trainable matrices**.

The original model weights remain unchanged.
Only the small LoRA layers are trained.

Why LoRA Is Powerful

LoRA offers several advantages:

- drastically fewer trainable parameters
- lower GPU memory usage
- faster training
- reusable base models

This allows many teams to fine-tune large models efficiently.

What Is QLoRA?

QLoRA extends LoRA by combining it with **model quantization**.

Quantization reduces **model precision to use less memory**.

QLoRA enables fine-tuning large models on **smaller hardware setups**.

When Developers Use LoRA / QLoRA

LoRA-based fine-tuning is useful when:

- domain knowledge is specialized
- consistent outputs are required
- prompts alone are insufficient
- custom model behavior is needed

Many open-model fine-tuning workflows rely on LoRA.

Key Takeaway

Modern fine-tuning is possible because of techniques like:

- LoRA
- QLoRA

These methods allow developers to adapt large models without retraining the entire network. Efficient fine-tuning makes custom AI models more accessible.

